

In one line

One complex quantity $\bar{\eta} = \rho + j\omega$ fuses the rate of change of a bus voltage's *magnitude* (ρ) and its *angular frequency* (ω), giving a single language for the dynamics, stability and synchronization of converter-dominated grids.

Key ideas

- **Complex frequency:** with $\bar{v} = v e^{u+j\theta}$ ($u = \ln v$),

$$\bar{\eta} = \frac{d}{dt}(u + j\theta) = \rho + j\omega, \quad \rho = \dot{v}/v, \quad \omega = \dot{\theta}.$$

- **Complex power** $\bar{s} = \bar{v} \circ \bar{i}^* = p + jq$, with rate of change (RoCoP) $\bar{s}' = \bar{\eta} \bar{s} + \bar{v} \circ \bar{i}'^*$.
- **Local synchronization:** a grid-side view of synchronism — device is synchronized when $\rho \rightarrow 0$ and $\omega \rightarrow \omega_r$ (bounded/asymptotic, BLS/ALS).
- **Transient Slack Capability (TSC):** what a device needs to sustain the grid after a large disturbance; realised by the **Dual grid-forming converter**.

Concept map

Park vectors $\rightarrow \bar{\eta}$ (complex freq.) $\rightarrow$ RoCoP $\bar{s}' \rightarrow$ device component $\bar{\chi} = \bar{\xi} - \bar{\eta} \rightarrow$ local sync (BLS/ALS) $\rightarrow$ TSC $\rightarrow$ Dual-GFM application.

Worked takeaways (verified)

- $\bar{v}' = \bar{v} \circ \bar{\eta}$; $\bar{i}' = \bar{\eta} \bar{i}$ (constant $\bar{Y}$).
- RoCoP: $\bar{s}' = \bar{\eta} \bar{s} + \bar{v} \circ \bar{i}'^*$.
- General component: $\bar{i}' = \bar{i}(\bar{\chi} + \bar{\eta}) \Rightarrow \bar{\chi} = \bar{\xi} - \bar{\eta}$.

Why ρ, ω combine: both are $[\text{time}]^{-1}$ (see `verify.md`).

Prerequisites (your profile)

Park/dq transform & Hadamard $\circ$ (`seen`); complex power (`followed`); target for $\bar{\eta}$ itself: `followed` $\rightarrow$ `intuitive`.

Where it fits

Milano's programme: *Complex Frequency* (TPWRS 2022, arXiv:2105.07769) $\rightarrow$ *Local Synchronization* (Ponce & Milano 2025) $\rightarrow$ *Dual-GFM & Transient Slack Capability* (2025).

Relevance to my work

For `effective_inertia`: the deck's TSC (port-Hamiltonian) separates stored energy H_k (Cond.1) from the dynamic slack to sustain ΔP (Cond.2) — structurally the same storage-vs-delivery split as your H_{load} vs H_{eff} / r (that framing is yours). $\bar{\eta} = \rho + j\omega$: you measure the ω (RoCoF) side — computing $\rho = \dot{v}/v$ too (you already log v_{rms}) could attribute your $\sim 0.2\text{s}$ offset to voltage vs frequency; ρ is also your CMLD voltage-response / feeder question. Low-inertia/IBR displacement + Dual-GFM motivate the setting.